

GROTHENDIECK TOPOLOGIES ON A FINITE CATEGORY FORM A SUBLATTICE OF THE LATTICE OF TWO-OUT-OF-THREE CLASSES

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ABSTRACT. For a small category S every Grothendieck topology J determines the class W_J of maps of presheaves inverted by sheafification, an element of the complete lattice $\text{Same}(\text{Presh}(S))$ of all classes of morphisms containing the identities and closed under composition and two-out-of-three. The map $J \mapsto W_J$ preserves all meets, and it was known to preserve binary joins for finite acyclic categories and for finite categories whose two topologies have an idempotent intersection of ideals, but not for the poset of natural numbers. We prove that it preserves binary joins for *every* finite category: $W_{J \vee J'} = W_J \vee W_{J'}$, the join on the right being the smallest class closed under composition and two-out-of-three containing both. The proof has two steps. Alternating separated quotients for the two topologies reach, in a bounded number of steps, the quotient of a presheaf by the equivalence relation of agreement on the join-covering sieves; and on such separated presheaves the alternating plus-construction tower is a chain of injections into the join sheafification, which it reaches because the join sheafification is, by an explicit retraction of the representing bimodules, a retract of every sufficiently late stage of the tower. The second step alone does not suffice: a monoid of order 8 is exhibited on which the alternating plus-construction tower never stabilizes, although the join is preserved.

1. INTRODUCTION

Let S be a small category, $\text{Top}(S)$ the lattice of Grothendieck topologies on S , and for $J \in \text{Top}(S)$ let a_J be the sheafification and $W_J = \{f : a_J f \text{ is an isomorphism}\}$ the class of J -local isomorphisms of presheaves [4, III.7]. A *sameness* on a category C is a class of morphisms containing the identities, closed under composition and under two-out-of-three; samenesses form a complete lattice $\text{Same}(C)$, whose joins are the smallest samenesses containing the union [6]. Each W_J is a sameness, $J \mapsto W_J$ is an order-embedding $\text{Top}(S) \hookrightarrow \text{Same}(\text{Presh}(S))$ preserving all meets, and always $W_J \vee W_{J'} \subseteq W_{J \vee J'}$ with equality iff the join on the left is saturated [8,

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Thms. 3.1, 4.2]. Equality holds for finite acyclic categories and for finite categories whose topologies have an idempotent intersection of ideals [8, Thms. 5.1, 6.10], and fails for the poset $\mathbb{N}$ [8, Thm. 7.1]; whether it holds for all finite categories was left open there and in [9, Question 6.8]. We answer this.

Theorem 1.1. *For every finite category S and all topologies J, J' on S ,*

$$W_J \vee W_{J'} = W_{J \vee J'} \quad \text{in } \text{Same}(\text{Presh}(S)).$$

Hence $J \mapsto W_J$ is a lattice embedding $\text{Top}(S) \hookrightarrow \text{Same}(\text{Presh}(S))$.

What is known. The finite-sieve dictionary identifies $\text{Top}(S)$, for finite S , with the idempotent two-sided ideals of S under reverse inclusion [7, Thm. 5.1]; the join $J_D \vee J_E$ corresponds to the largest idempotent ideal I' contained in $D \cap E$. The *reduction lemma* [8, Lemma 4.3] shows that $W_D \vee W_E = W_{I'}$ as soon as the alternating tower of plus constructions $X \rightarrow X^{+D} \rightarrow X^{+D+E} \rightarrow \dots$ reaches an I' -sheaf for every presheaf X , and the two positive results above were proved by showing that it does. The tower is represented by the alternating tensor tower $T_k = D_k \otimes \dots \otimes D_1$ of bimodules ($\Phi_k X = \text{Hom}(T_k, X)$); in [9, §6.3] it is shown that for every finite category $(T_k)_k$ is pro-isomorphic to $I' \otimes I'$, the bimodule representing $a_{I'}$. The present paper does not use that result: the only consequence needed, that $a_{I'} X$ is a retract of the late stages of the tower, is proved here directly by an explicit retraction (Lemma 2.3).

Related work. The lattice of subtoposes, or of local operators, of a presheaf topos is classical [4, 2]; for essential localizations it is described by Kelly and Lawvere through idempotent ideals [3], and every Grothendieck topology on a finite Cauchy-complete category is rigid [5], which is the background of the object-generated ideals used below. For ∞ -topoi, Anel, Biedermann, Finster and Joyal describe left-exact localizations by Grothendieck topologies [1]. All of this concerns the lattice of *saturated* classes, i.e. of the localizations themselves. The question here is different: after associating to each topology its class of local isomorphisms, does the bare two-out-of-three join, before any saturation, already coincide with the class of local isomorphisms of the joined topology? For finite categories the answer is yes, and the proof shows which maps outside the canonical plus-construction units are needed.

What is new. Theorem 1.1, and the mechanism of its proof. The plus tower alone does not always stabilize: Section 7 gives a monoid of order 8 on which it never does. The missing ingredient is a second, dual tower of *separated quotients*, each of which identifies the elements of a presheaf that agree on one of the two topologies. Three facts combine: local isomorphisms for a topology whose ideal is generated by objects are the maps bijective at those objects (Lemma 3.1); the alternating separated quotients reach, in a bounded number of steps, the quotient by agreement on the join-covering

sieves (Lemma 4.2); and on a presheaf separated for the join topology, plus constructions preserve separatedness, so the plus tower is a chain of injections into the join sheafification $a_{I'}X$, which it reaches because $a_{I'}X$ is a retract of every stage beyond $4m$ (Lemmas 2.3 and 5.1). The proofs are self-contained.

2. PRELIMINARIES

Throughout, S is a finite category. Presheaves are functors $S^{\text{op}} \rightarrow \mathbf{Set}$; for $a: x \rightarrow c$ and $v \in X(c)$ we write $v \cdot a := X(a)v \in X(x)$. A *two-sided ideal* of S is a class of morphisms closed under composition with arbitrary morphisms on either side; it is *idempotent* if every member is a composite of two members. By the dictionary [7, Thm. 5.1], topologies on S correspond to idempotent ideals: J_D has as covering sieves on c the sieves containing $D(-, c) = \{a: x \rightarrow c \text{ in } D\}$, and $J_D \leq J_E$ iff $D \supseteq E$. For the plus construction and sheafification of J_D we have $X^+(c) = \text{Hom}(D(-, c), X)$ (maps of presheaves from the sieve) and $a_DX = X^{++}$ [4, III.5].

2.1. Cauchy completion and generating objects. Every element d of an idempotent ideal D of a finite category is a composite $d = d_N \cdots d_1$ of members for every N ; for N larger than the number of morphisms two partial composites $d_i \cdots d_1, d_j \cdots d_1$ ($i < j$) coincide, so $d_i \cdots d_1 = w \cdot d_i \cdots d_1$ for the endomorphism $w = d_j \cdots d_{i+1} \in D$, and a power $g = w^r$ is an idempotent of D with $d = (d_N \cdots d_{j+1})g(d_i \cdots d_1)$. Hence D is the ideal generated by the idempotent endomorphisms it contains. Passing to the Karoubi envelope $\mathcal{K}$ of S (idempotents split; presheaf categories are equivalent, and topologies, local isomorphisms and sheafifications correspond), every idempotent ideal D becomes the ideal generated by the set of objects

$$\mathcal{D} := \{d \in \text{Ob } \mathcal{K} : \text{id}_d \in D\},$$

in the sense that $D(-, c)$ is the set of morphisms into c factoring through an object of $\mathcal{D}$. We may therefore, and do, assume that S is Cauchy complete and that D, E are the ideals generated by the object sets $\mathcal{D} = \{d : \text{id}_d \in D\}$ and $\mathcal{E} = \{e : \text{id}_e \in E\}$; the join $J_D \vee J_E$ is then $J_{I'}$ where I' is the ideal generated by $\mathcal{J} := \mathcal{D} \cap \mathcal{E} = \{j : \text{id}_j \in D \cap E\}$ (the idempotent ideal generated by the idempotents of $D \cap E$ is the largest idempotent ideal in $D \cap E$). Each of the descending chains $(DE)^r$ and $(ED)^r$, $r \geq 1$, stabilizes at an idempotent ideal contained in $D \cap E$ and containing $I' = I'^{2r}$, hence at I' ; we fix $m \geq 1$ with $(DE)^m = (ED)^m = I'$, so that every product of $2m$ alternating factors D, E , in either order, equals I' .

2.2. Samenesses and the join theorem. For a functor P write $W_P = \{f : Pf \text{ invertible}\}$, a sameness.

Theorem 2.1 (join theorem, [7, Thm. 4.1]). *Let U, V be samenesses on C whose join contains the isomorphisms of C (as is the case when U or V is of the form W_F), and let $P: C \rightarrow C$ be a functor with a natural transformation*

$\pi: \text{id} \Rightarrow P$ such that $\pi_X \in U \vee V$ for all X and P inverts U and V . Then $U \vee V = W_P$.

Proof. $W_P \supseteq U \vee V$ since W_P is a sameness containing U and V . If Pf is invertible, then $Pf \in U \vee V$ by hypothesis, so the naturality square $\pi_Y f = Pf \pi_X$ has three sides in $U \vee V$, hence the fourth. $\square$

2.3. The plus tower and its representing bimodules. An S – S -bimodule is a functor $S^{\text{op}} \times S \rightarrow \mathbf{Set}$; ideals are sub-bimodules of Hom_S , and for bimodules A, B the tensor $(A \otimes B)(x, z) = \coprod_y A(y, z) \times B(x, y) / ((a \cdot m, b) \sim (a, m \cdot b))$ is a bimodule. For a bimodule A and a presheaf X the presheaf $\text{Hom}(A, X)(c) := \text{Hom}_{\text{Presheaf}(S)}(A(-, c), X)$ satisfies the adjunction $\text{Hom}(A \otimes B, X) \cong \text{Hom}(A, \text{Hom}(B, X))$ and, for $A = \text{Hom}_S$, $\text{Hom}(\text{Hom}_S, X) \cong X$; a map of bimodules $\alpha: A \rightarrow A'$ induces $\alpha^*: \text{Hom}(A', X) \rightarrow \text{Hom}(A, X)$, and by the Yoneda lemma α is an isomorphism iff α^* is bijective for all X . Elements of $(A_k \otimes \cdots \otimes A_1)(x, c)$ are classes $[a_k, \dots, a_1]$ of composable chains $x \rightarrow \cdots \rightarrow c$ with $a_i \in A_i$, modulo moving a morphism of S across a tensor sign.

Write $D_1 = D$, $D_2 = E$, $D_3 = D, \dots$. Since $D(-, c)$ is the least J_D -covering sieve, $X^{+D} = \text{Hom}(D, X)$, with unit $x \mapsto (d \mapsto x \cdot d)$ induced by the multiplication $D \rightarrow \text{Hom}_S$. Hence the alternating plus tower is

$$\Phi_k X := X^{+D_1+D_2+\cdots+D_k} = \text{Hom}(T_k, X), \quad T_k := D_k \otimes \cdots \otimes D_1,$$

with unit $u_k: X \rightarrow \Phi_k X$, $u_k(x) = (t \mapsto x \cdot \bar{t})$, where $\bar{t} = d_k \cdots d_1$ is the composite of the chain t ; u_k is a composite of plus-construction units, hence of maps of $W_D \cup W_E$. Likewise $a_{I'} X = X^{+I'+I'} = \text{Hom}(P, X)$ for $P := I' \otimes I'$, with unit $\eta_X(x) = (p \mapsto x \cdot \bar{p})$.

Lemma 2.2. *Let $m_k: P \otimes T_k \rightarrow P$ be the bimodule map $[i, i', d_k, \dots, d_1] \mapsto [i, i' d_k \cdots d_1]$ (well defined since I' is a right ideal). Then m_k is an isomorphism of bimodules, and $m_k^* = a_{I'}(u_k): a_{I'} X \rightarrow a_{I'} \Phi_k X$ under the identification $a_{I'} \Phi_k X = \text{Hom}(P, \text{Hom}(T_k, X)) = \text{Hom}(P \otimes T_k, X)$.*

Proof. For $\psi \in a_{I'} X = \text{Hom}(P, X)$ the map $a_{I'}(u_k)(\psi) = u_k \circ \psi$ sends p to $(t \mapsto \psi(p) \cdot \bar{t}) = (t \mapsto \psi(p \cdot \bar{t}))$, i.e. corresponds to $[p, t] \mapsto \psi(m_k[p, t])$; so $a_{I'}(u_k) = m_k^*$. Since $u_k \in W_D \vee W_E \subseteq W_{I'}$ ($J_D, J_E \leq J_{I'}$), $a_{I'}(u_k)$ is bijective for every X , and m_k is an isomorphism by Yoneda. $\square$

Let $\iota_k: \Phi_k X \rightarrow a_{I'} X$ be the sheafification unit of $\Phi_k X$ followed by $(a_{I'} u_k)^{-1}$; thus $\iota_k = (m_k^*)^{-1} \circ \varepsilon_k^*$ where $\varepsilon_k: P \otimes T_k \rightarrow T_k$, $[p, t] \mapsto \bar{p} \cdot t$, is the bimodule map inducing the unit $\Phi_k X \rightarrow a_{I'} \Phi_k X$. By naturality of units, $\iota_k u_k = \eta_X$.

Lemma 2.3 (retraction). *Let $k \geq 4m$ and choose j with $2m \leq j \leq k - 2m$. The map*

$$r_k: T_k \rightarrow P, \quad [d_k, \dots, d_1] \mapsto [d_k \cdots d_{j+1}, d_j \cdots d_1],$$

is a well-defined map of bimodules (both products lie in I' , being products of at least $2m$ alternating factors), and $r_k \circ \varepsilon_k = m_k$. Consequently $\sigma_k :=$

$r_k^*: a_{I'}X \rightarrow \Phi_k X$ satisfies $\iota_k \sigma_k = \text{id}$; in particular ι_k is surjective for every presheaf X and every $k \geq 4m$.

Proof. A tensor move inside one of the two blocks does not change its product; a move across the boundary, $[\dots, d_{j+1}m, d_j, \dots] \sim [\dots, d_{j+1}, md_j, \dots]$, changes the pair of products from (Am, B) to (A, mB) , which are identified in $P = I' \otimes I'$. So r_k is well defined, and it is clearly a bimodule map. For $[p, t] \in P \otimes T_k$ with $p = [i, i']$, $r_k \varepsilon_k[p, t] = [ii'd_k \cdots d_{j+1}, d_j \cdots d_1] = [i, i'd_k \cdots d_1] = m_k[p, t]$, the middle equality being a tensor move in P (both i and $d_j \cdots d_1$ lie in I'). Hence $\iota_k \sigma_k = (m_k^*)^{-1} \varepsilon_k^* r_k^* = (m_k^*)^{-1} (r_k \varepsilon_k)^* = \text{id}$. $\square$

3. LOCAL ISOMORPHISMS ARE OBJECTWISE BIJECTIONS

Lemma 3.1. *Let D be the ideal generated by a set of objects $\mathcal{D}$ of the Cauchy complete category S . A map of presheaves $f: X \rightarrow Y$ is a J_D -local isomorphism iff $f_d: X(d) \rightarrow Y(d)$ is bijective for every $d \in \mathcal{D}$. Consequently $W_D = \{f : f \text{ bijective at } \mathcal{D}\}$, $W_E = \{f : f \text{ bijective at } \mathcal{E}\}$ and $W_{I'} = \{f : f \text{ bijective at } \mathcal{J}\}$.*

Proof. f is J_D -locally injective iff for all c and $x, x' \in X(c)$ with $fx = fx'$ the sieve $\{a : x \cdot a = x' \cdot a\}$ contains $D(-, c)$, and J_D -locally surjective iff for all $y \in Y(c)$ the sieve $\{a : y \cdot a \in \text{im} f\}$ contains $D(-, c)$ [4, III.7]. Every $a \in D(-, c)$ factors as $a = p \circ q$ with $p: d \rightarrow c$, $d \in \mathcal{D}$. If f_d is injective, then $f(x \cdot p) = (fx) \cdot p = (fx') \cdot p = f(x' \cdot p)$ gives $x \cdot p = x' \cdot p$, hence $x \cdot a = x' \cdot a$; conversely local injectivity with $c = d$ and $a = \text{id}_d$ gives injectivity at d . If f_d is surjective then $y \cdot p = f(x)$ for some $x \in X(d)$ and $y \cdot a = f(x \cdot q)$; conversely local surjectivity with $a = \text{id}_d$ gives surjectivity at d . The last assertion uses that I' is generated by $\mathcal{J}$. $\square$

For a finite monoid M and an idempotent two-sided ideal D : a map of right M -sets f is a J_D -local isomorphism iff $f|_{Xe}: Xe \rightarrow Ye$ is bijective for every idempotent $e \in D$.

4. SEPARATED QUOTIENTS

Definition 4.1. For a presheaf X and an ideal D generated by objects $\mathcal{D}$, let $x \approx_D x'$ (for $x, x' \in X(c)$) iff $x \cdot a = x' \cdot a$ for all $a \in D(-, c)$; equivalently, iff $x \cdot p = x' \cdot p$ for all $p: d \rightarrow c$ with $d \in \mathcal{D}$. The D -separated quotient is $Q_D X := X / \approx_D$.

If $x \approx_D x'$ then $x \cdot b \approx_D x' \cdot b$ for every $b: c' \rightarrow c$ (as $(x \cdot b) \cdot a = x \cdot (ba)$ and $ba \in D(-, c)$), so $Q_D X$ is a presheaf and $q: X \twoheadrightarrow Q_D X$ is a map; $Q_D X$ is J_D -separated, and q is bijective at every $d \in \mathcal{D}$ (take $a = \text{id}_d$), so $q \in W_D$ by Lemma 3.1.

Lemma 4.2. *Let $X_0 = X$ and $X_{k+1} = Q_{D_{k+1}} X_k$ with the alternation $D_1 = D$, $D_2 = E, \dots$. Then two elements of $X(c)$ have the same image in X_k iff they agree on restriction along every morphism of the product ideal $D_k \circ \cdots \circ D_1$; in particular $X_k = Q_{I'} X$ for $k \geq 2m$, where $Q_{I'} X$ is the quotient by*

Proof. Induction on k : $x, x' \in X(c)$ have the same image in X_{k+1} iff $x \cdot b$ and $x' \cdot b$ have the same image in X_k for all $b \in D_{k+1}(-, c)$, i.e. iff $x \cdot (ba) = x' \cdot (ba)$ for all such b and all $a \in (D_k \cdots D_1)(-, \text{dom } b)$; the morphisms ba are exactly those of $D_{k+1}D_k \cdots D_1$. For $k \geq 2m$ this ideal is I' , and agreement along I' is agreement along the morphisms $p: j \rightarrow c$ with $j \in \mathcal{J}$ (every $a \in I'(-, c)$ is $p \circ q$ with such a p). Each quotient map $X_k \rightarrow X_{k+1}$ lies in $W_{D_{k+1}}$. $\square$

Call a presheaf $\mathcal{F}$ -*separated* if it is $J_{I'}$ -separated, i.e. $x \cdot p = x' \cdot p$ for all $p: j \rightarrow c$, $j \in \mathcal{F}$, implies $x = x'$.

Lemma 5.1. *Let Y be $\mathcal{J}$ -separated and $\mathcal{J} \subseteq \mathcal{D}$. Then Y^{+D} is $\mathcal{J}$ -separated. Consequently every stage $\Phi_k Y$ of the alternating plus tower is $\mathcal{J}$ -separated, the maps $\iota_k: \Phi_k Y \rightarrow a_I Y$ are injective, and for $k \geq 4m$ they are bijective; the unit $Y \rightarrow a_I Y$ lies in $W_D \vee W_E$.*

Proof. For $j \in \mathcal{J} \subseteq \mathcal{D}$ the sieve $D(-, j)$ is maximal, so $Y^{+D}(j) = Y(j)$, and for $\varphi \in Y^{+D}(c) = \text{Hom}(D(-, c), Y)$ and $p: j \rightarrow c$ the restriction is $\varphi \cdot p = \varphi(p)$. Let $\varphi, \psi \in Y^{+D}(c)$ satisfy $\varphi(p) = \psi(p)$ for all $p: j \rightarrow c$ with $j \in \mathcal{J}$. For any $n: x \rightarrow c$ in $D(-, c)$ and any $b: j \rightarrow x$ with $j \in \mathcal{J}$, $\varphi(n) \cdot b = \varphi(nb) = \psi(nb) = \psi(n) \cdot b$, and Y is $\mathcal{J}$ -separated, so $\varphi(n) = \psi(n)$; thus $\varphi = \psi$. The same holds for E . Hence all $\Phi_k Y$ are $\mathcal{J}$ -separated. The map ι_k is the sheafification unit of $\Phi_k Y$ followed by an isomorphism (Section 2.3), and the unit of a separated presheaf is injective [4, III.5]; so ι_k is injective. By Lemma 2.3, ι_k is surjective for $k \geq 4m$. So ι_k is an isomorphism; isomorphisms lie in $W_D \cap W_E \subseteq W_D \vee W_E$ (a sameness need only contain the identities, but W_D and W_E are classes of maps inverted by a functor), so the unit $Y \rightarrow a_{I'} Y = \iota_k u_k$ lies in $W_D \vee W_E$. $\square$

$$\begin{array}{ccccccc}
X & \xrightarrow{W_D} \twoheadrightarrow & Q_D X & \xrightarrow{W_E} \twoheadrightarrow & \cdots & \longrightarrow \twoheadrightarrow & Q_{I'} X^{u_k} \xrightarrow{\in W_D \vee W_E} \Phi_k Q_{I'} X \\
& & & & & & \downarrow \simeq \downarrow \iota_k \\
& & & & & & a_{I'} X
\end{array}$$

FIGURE 1. The proof of Theorem 1.1: separated quotients (at most $2m$) followed by plus constructions ($4m$) connect every presheaf to its join sheafification inside the join.

6. PROOF OF THE THEOREM

Proof of Theorem 1.1. By Section 2 we may assume S Cauchy complete, $J = J_D$, $J' = J_E$ with D, E generated by objects, and $J \vee J' = J_{I'}$. Apply the join theorem with $U = W_D$, $V = W_E$, $P = a_{I'}$ and π the unit $\eta: \text{id} \Rightarrow a_{I'}$. P inverts W_D and W_E since $J_D, J_E \leq J_{I'}$. For a presheaf X the unit factors as

$$X \twoheadrightarrow Q_{I'} X \xrightarrow{\eta} a_{I'} Q_{I'} X \cong a_{I'} X,$$

the first map in the join by Lemma 4.2, the second by Lemma 5.1 ($Q_{I'} X$ is $\mathcal{J}$ -separated), and the isomorphism induced by the quotient map, which $a_{I'}$ inverts. Hence $\eta_X \in W_D \vee W_E$ for all X , and $W_D \vee W_E = W_{a_{I'}} = W_{I'}$. Since $W_{J \vee J'} = \text{Sat}(W_J \vee W_{J'}) \supseteq W_J \vee W_{J'}$ always, and $J \mapsto W_J$ preserves meets [8, Thm. 3.1], the map is a lattice embedding. $\square$

Remark 6.1. The number of steps is uniform: $2m$ quotients and $4m$ plus constructions, depending only on (S, D, E) . The proof uses finiteness of S only through the existence of m , the stabilization of the alternating products of the two ideals at I' ; this fails for $\mathbb{N}$, where the join is not preserved [8, Thm. 7.1]. The number $2m$ of quotients is also the number of alternating iterations of the two Lawvere–Tierney local operators needed to reach their join, since $j_D j_E(R) = \{a : a \cdot DE \subseteq R\}$ on sieves R .

Remark 6.2. Nothing in the argument requires S to be Cauchy complete except the description of ideals by objects; for a finite category S the statement is transported along the equivalence $\text{Presh}(S) \simeq \text{Presh}(\mathcal{K})$.

7. THE PLUS TOWER ALONE DOES NOT SUFFICE

The reduction lemma of [8] proves join preservation whenever the alternating plus tower reaches an I' -sheaf. The following example shows that Lemma 4.2 cannot be dispensed with.

Proposition 7.1. *Let $\overline{M} = \{1, 1_a, 1_b, u, v, uv, vu, 0\}$ be the contracted monoid of the category with two objects a, b , arrows $u: a \rightarrow b$, $v: b \rightarrow a$, in which all composites of length ≥ 3 are identified with 0; $|\overline{M}| = 8$. Let D be the ideal generated by 1_b and E the ideal generated by 1_a ; then $D \cap E = \{u, v, uv, vu, 0\}$ is not idempotent and $I' = \{0\}$. The alternating plus-construction tower of (D, E) never stabilizes: for every even k the composite multiplication map $T_{k+4} \rightarrow T_k$ of the representing tensor tower is not surjective. In particular the maps $T_{k+1} \rightarrow T_k$ are not eventually bijective, and there is no k with $\Phi_k X$ an I' -sheaf for all X . Nevertheless $W_D \vee W_E = W_{I'}$ by Theorem 1.1.*

Proof. Here $m = 2$, so the image of the composite $T_{k+4} \rightarrow T_k$ is $(D_{k+4} \cdots D_{k+1}) \cdot T_k = I' \cdot T_k$ (Section 2), the set of classes with a representative whose first slot lies in $I' = \{0\}$. We show that $T_k \neq I' \cdot T_k$ for every even k ; the last two assertions then follow, since bijectivity of μ_{k+1} for all $k \geq K$ would make $T_{K+4} \rightarrow T_K$ bijective, and since, if

$\Phi_k X$ were an I' -sheaf for all X , then $\Phi_k X$ would be a D -sheaf and an E -sheaf ($J_D, J_E \leq J_{I'}$), all later plus-construction units on it would be isomorphisms and $\Phi_k X \rightarrow \Phi_{k+l} X$ would be bijective for all l and X : for even k this makes $T_{k+4} \rightarrow T_k$ an isomorphism of bimodules by Yoneda, and for odd k it makes $\Phi_{k+1} X \cong \Phi_k X$ an I' -sheaf for all X with $k+1$ even. Consider, for even $k = 2j$, the class of $t = (1_a, uv, 1_a, uv, \dots) \in T_k$, whose slots alternate between E and D as required. Tuples of non-zero elements with product the word $w = (uv)^j$ of the free category are *cut configurations* of w : cut points $0 = c_0 \leq \dots \leq c_k = 2j$, the i -th slot being the subword between c_{i-1} and c_i , an empty slot being the identity at its cut point, which must be a (even cut point) for an E -slot (i odd) and b (odd cut point) for a D -slot (i even); the slots have length ≤ 2 , since longer words are 0. A tensor move between tuples of non-zero elements moves one cut point. Put $d_i = i - c_i$; a slot of length ℓ changes d by $1 - \ell$, and an empty slot raises d by one and is allowed only from an even value of d (the parity condition on c_i). From an odd value d cannot rise, so $d \leq 1$ throughout; a slot of length ≥ 3 would send d to a value ≤ -1 , from which $d_k = 0$ can no longer be reached. Hence no cut configuration has a slot of length ≥ 3 , and no tensor move applied to a cut configuration produces a zero slot (a zero would have to arise as a composite of length ≥ 3 or as a non-composable pair, and adjacent slots of a configuration are composable). So the class of t consists of cut configurations only and has no representative with a zero slot; thus $T_k \neq I' \cdot T_k$ for every even k . $\square$

Remark 7.2. Enumerating the tensor relations gives $|T_1| = 6$ and $|T_k| = 5$ for $k \geq 2$: besides the zero class, four classes of cut configurations of non-zero words persist at every level, and no multiplication map is surjective.

By contrast, the separated quotients collapse these classes at once: on $\overline{M}$ -sets the quotient Q_E identifies two elements that agree under $1_a, u, v, uv, vu$ and 0, and since $m = 2$ here, four alternating quotients produce the I' -separated quotient, which is already the I' -sheaf: the set of elements fixed by 0. The derivation of the unit $X \rightarrow a_{I'} X$ inside $W_D \vee W_E$ is thus a composite of quotient maps, none of which is a plus-construction unit.

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